Kathará

Laboratory: OSPF multiarea

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What are OSPF areas?

- An abstraction that **simplifies administration** and **improves scalability**
  - the topology of an area is invisible from the outside
  - routers internal to a given area don't see the detailed external topology
- Each area runs a separate instance of the link state routing algorithm
  - all routers in an area construct the same lsdb
  - each router keeps **a distinct lsdb for each area it belongs to**

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OSPF areas

- They are identified by a 32-bit number, often in dotted decimal notation (1.2.3.4)
  - different interfaces of the same router can be assigned to different areas
  - each
    - router interface...
    - network...
    - router adjacency...

    ...is associated with a single area

Area types

- backbone (0.0.0.0)
  - **must be (virtually) connected**
  - all other areas are connected to it
  - contains all the area border routers
- stub
  - It does not receive advertisements of external routes (?)
  - **internal nodes are offered a default route**
  - cannot contain autonomous system boundary routers
  - the backbone can’t be a stub area
- transit
  - used to pass traffic from one adjacent area to another, via virtual links

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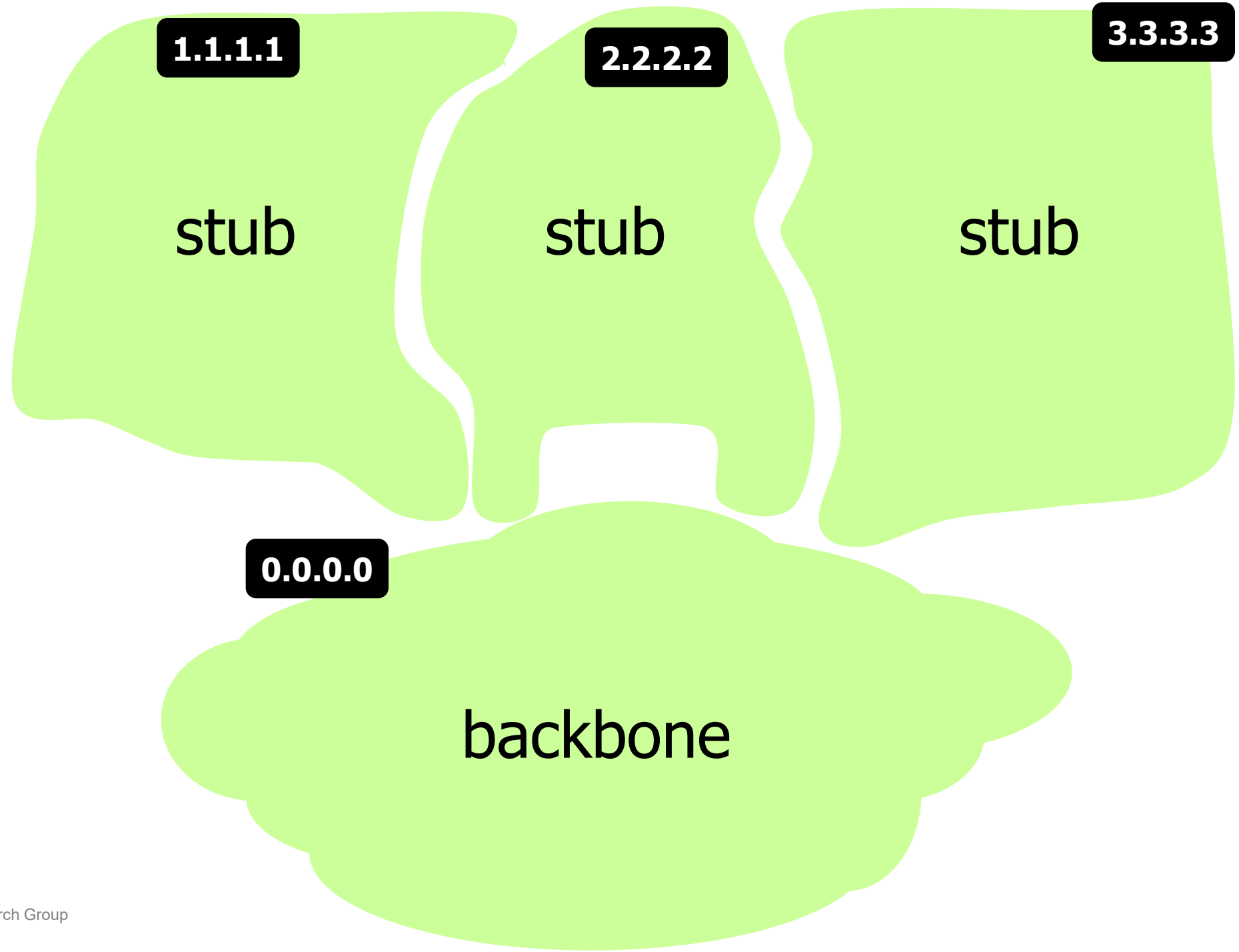

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Router types

- **internal** router
  - all interfaces belong to the same area
- **area border** router (abr)
  - connects one or more areas to the backbone
  - **keeps multiple lsdbs**, one for each area
- **backbone** router
  - has at least one interface connected to the backbone
  - **an abr is always a backbone router**
- **autonomous system boundary** router (asbr)
  - imports and floods routing information from other routing protocols (typically, bgp)
- note: **a router can be of more than one type**

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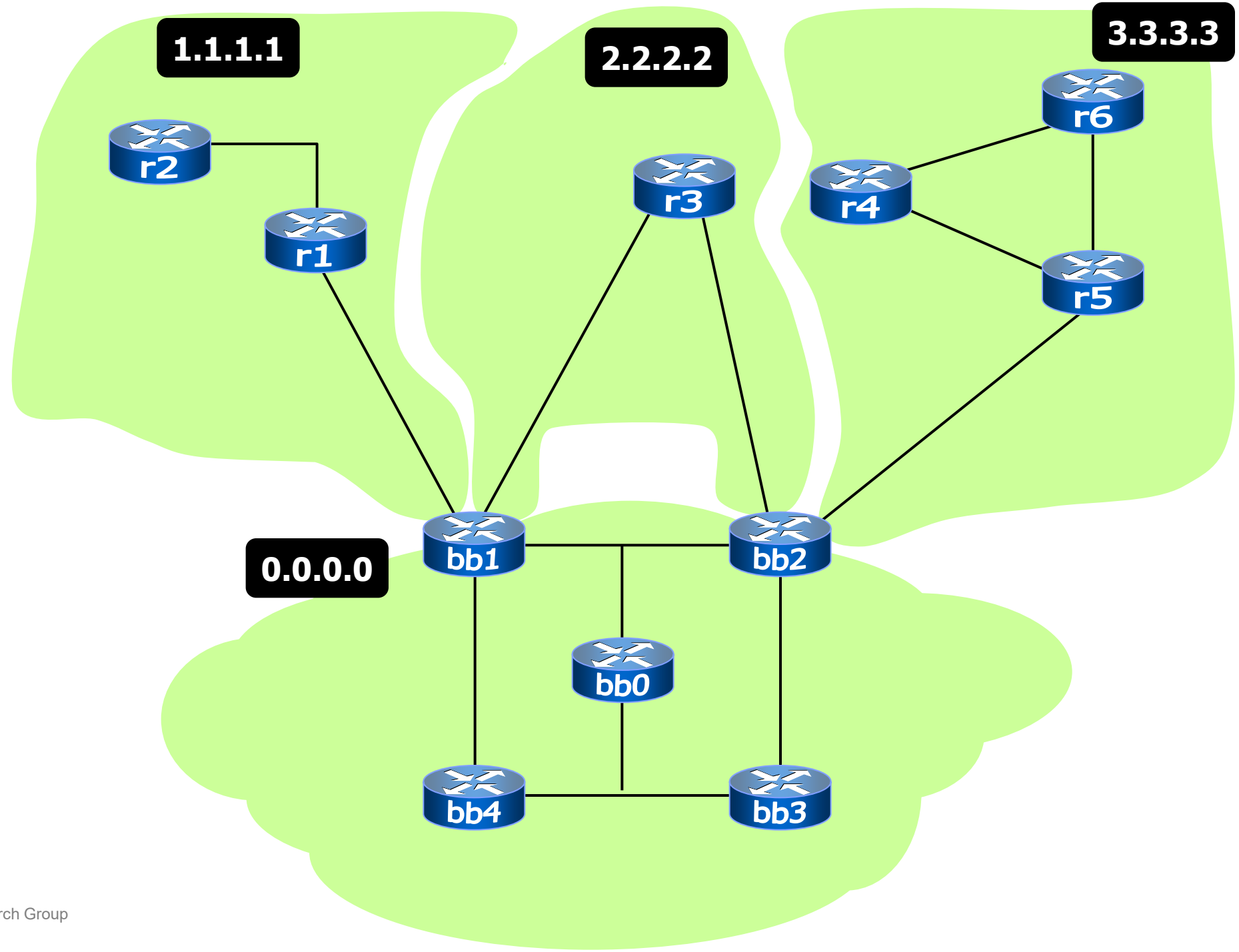

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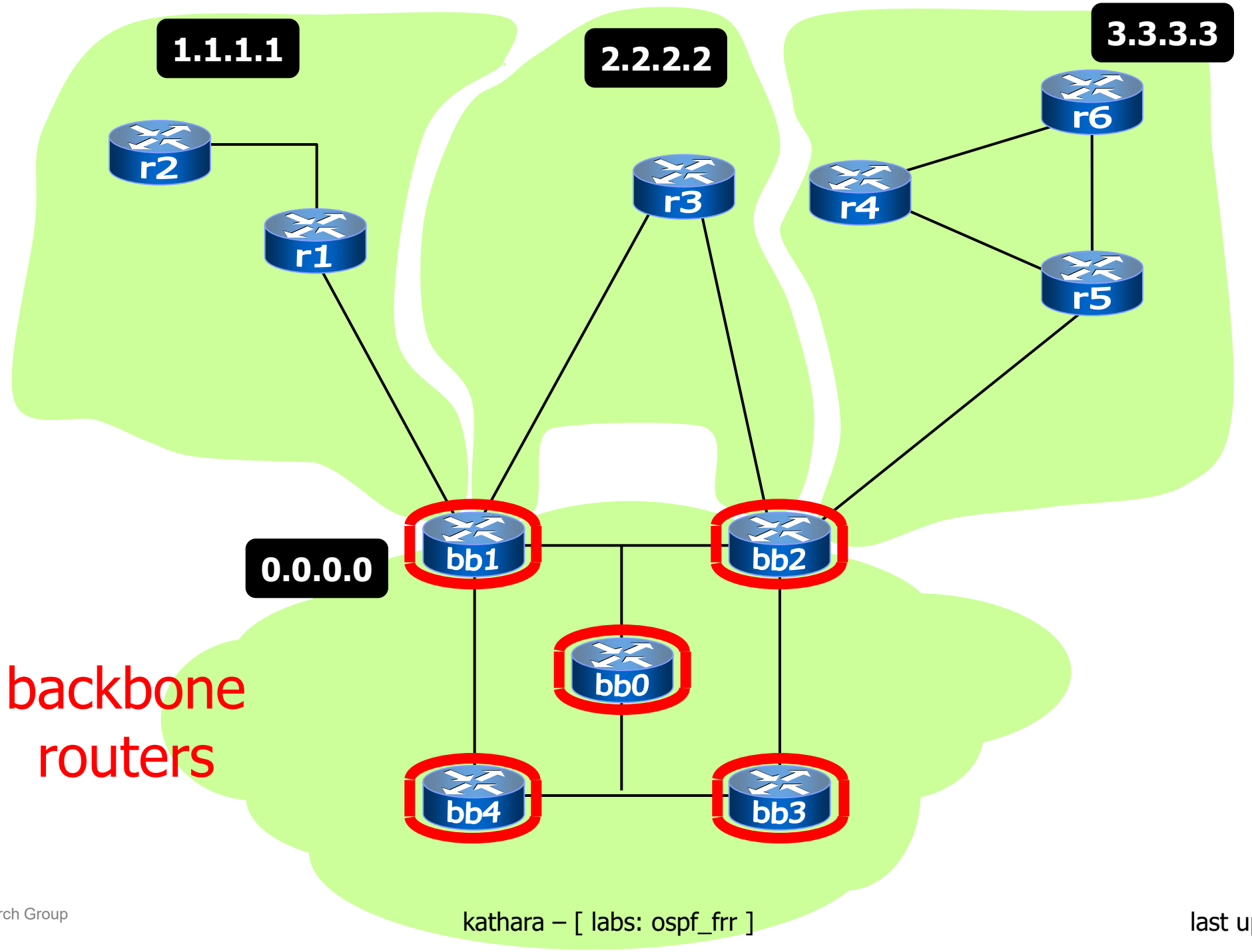

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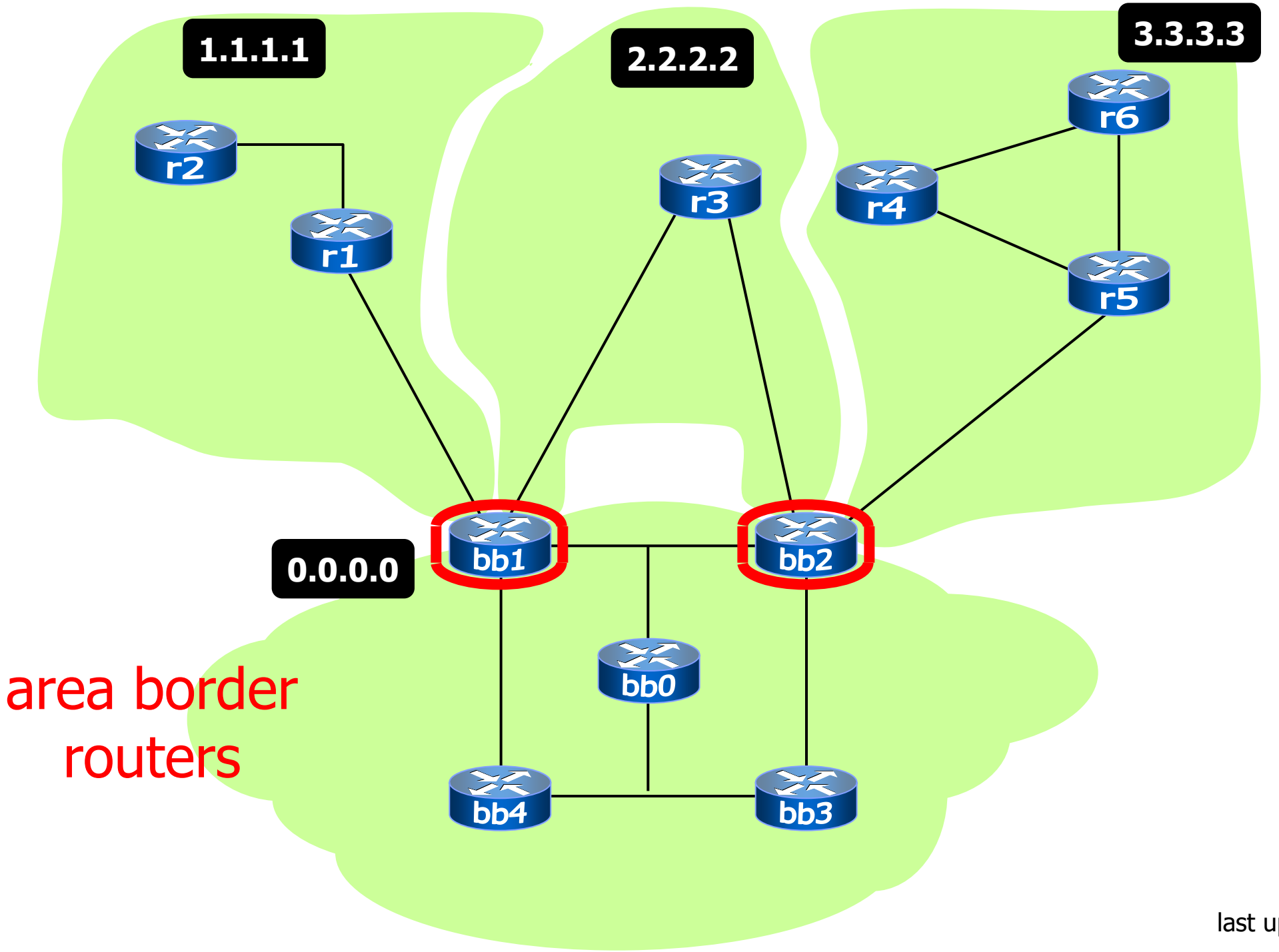

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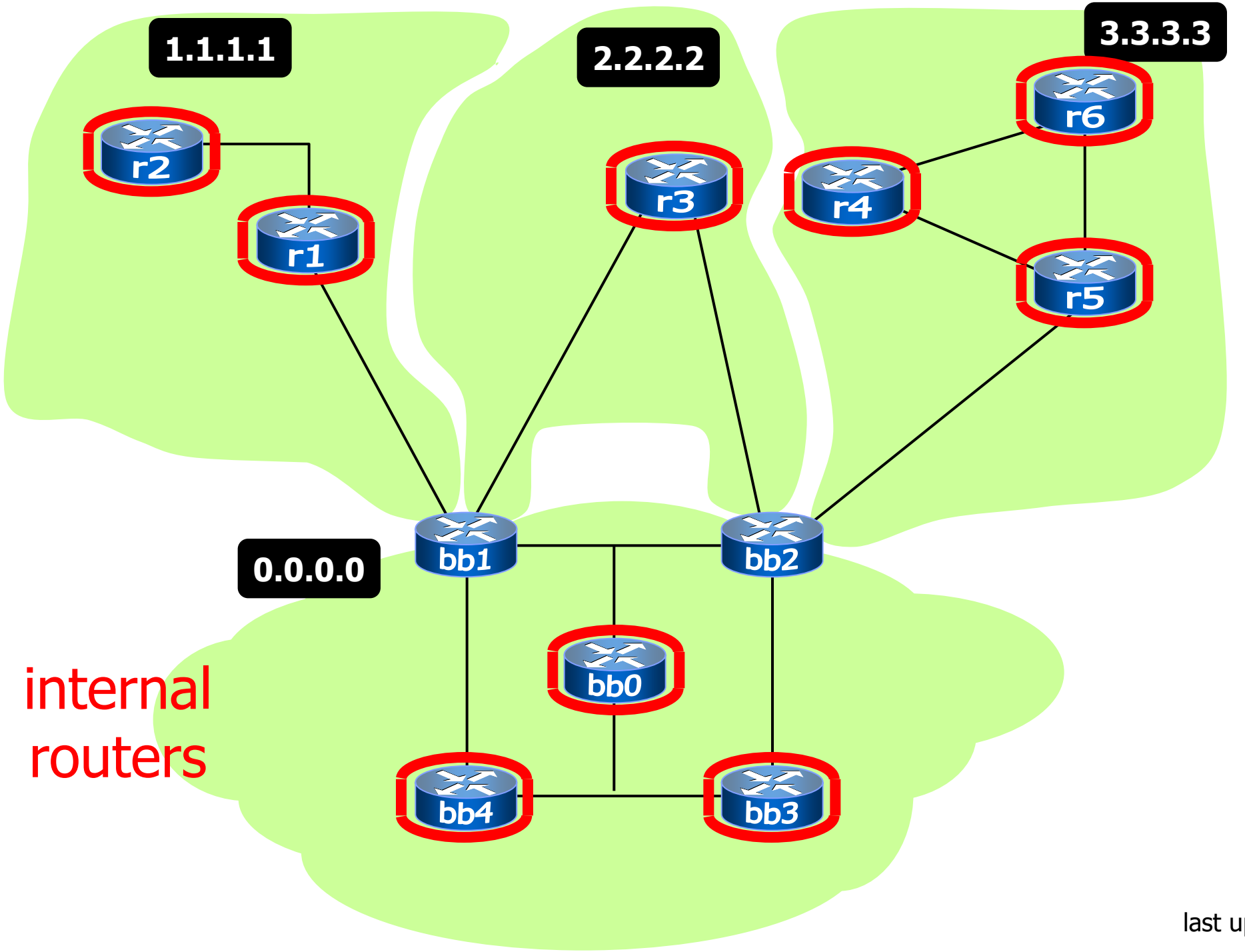

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Path Types

- OSPF classifies routes into path types based on their origin and location relative to areas,
- strict preference order for shortest path selection:
- intra-area > inter-area > external type 1 (E1) > external type 2 (E2).

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Path types

- Intra-Area (O)

  Routes to destinations within the same OSPF area, using the full link-state database for SPF calculation. Highest preference as they reflect local topology details.

- Inter-Area (O IA)

  Routes to other OSPF areas, learned via ABR summary LSAs (Type 3). Cost includes intra-area to ABR + backbone path + intra-area from destination ABR; visible as aggregated summaries.

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Path types

- External Type 1 (E1)

  Redistributed external routes (e.g., from BGP, static) where total cost = external metric + OSPF internal cost to ASBR.

  Preferred over E2; used when internal paths vary significantly.

- External Type 2 (E2)

  Default for external routes; cost = external metric only (OSPF cost to ASBR as tie-breaker).

  Ideal when external costs dominate or ASBRs are equidistant.

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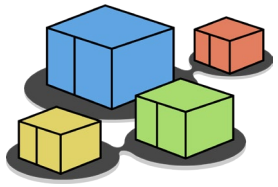
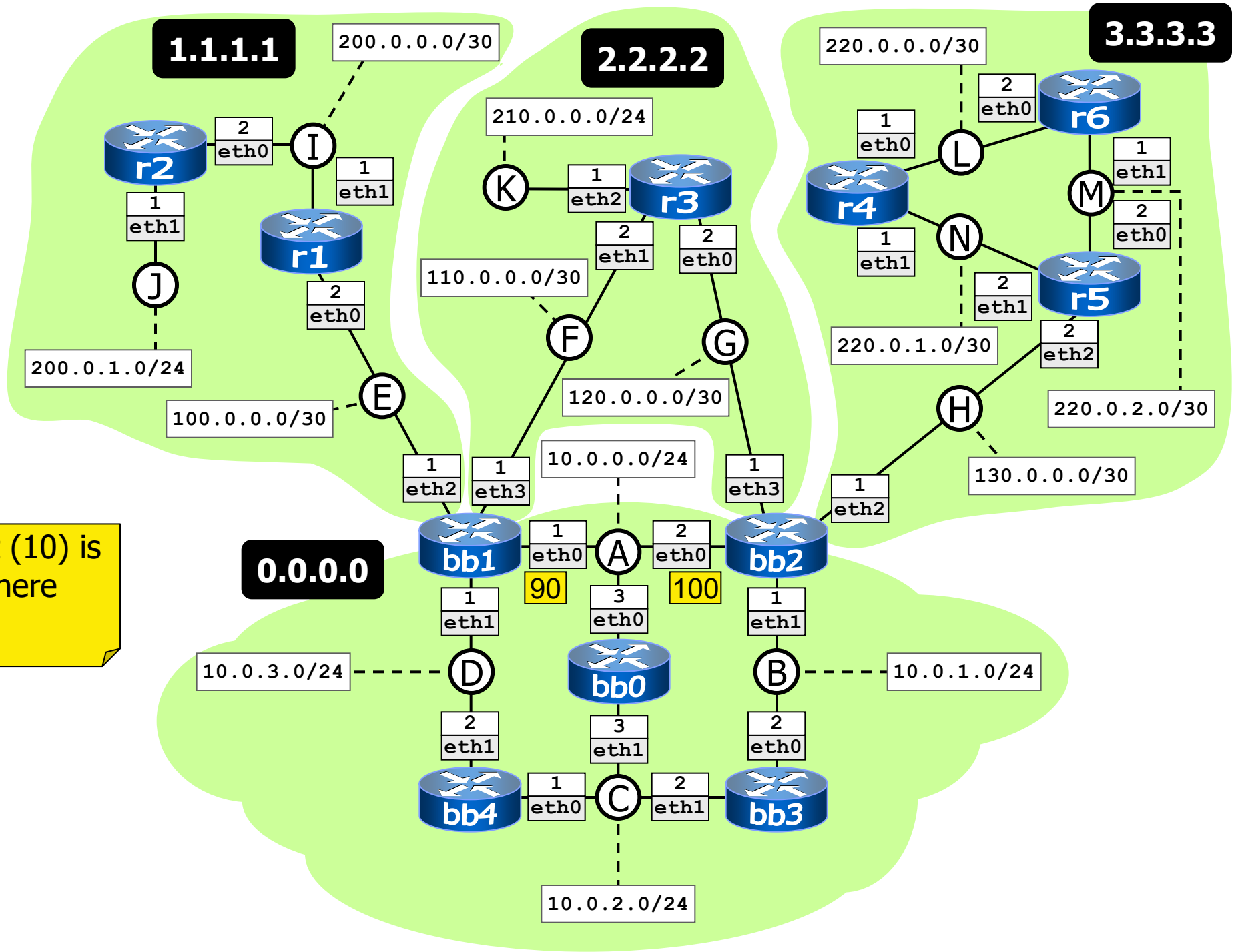


default cost (10) is assumed where unspecified



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area configuration

- area information is found in two places
  - when enabling ospf on router interfaces
    `network 200.0.0.0/16 area 1.1.1.1`
  - when specifying the area type (not required for the backbone)
    `area 1.1.1.1 stub`

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```
!
! FRRouting configuration file
!
!
!  OSPF CONFIGURATION
!
interface eth0
ospf cost 90
!
router ospf
! Speak OSPF on all interfaces falling in the listed subnets
network 10.0.0.0/16 area 0.0.0.0
network 100.0.0.0/30 area 1.1.1.1
network 110.0.0.0/30 area 2.2.2.2
area 1.1.1.1 stub
area 2.2.2.2 stub
!
!
log file /var/log/frr/frr.log
```

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ospf path types

- there are 4 path types
  1. intra-area
  2. inter-area
  3. external type 1
  4. external type 2
- types can coexist in the same network
- each type is preferred over the following ones

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ospf path types

- intra-area paths
  - calculated using the shortest-path tree

ospf path types

- inter-area paths
  - **abrs inject summary information inside each area**, to make it aware of available destinations in other areas
    - such information includes the **cost of the shortest path from the abr to the destination**
    - if multiple subnets are summarized into a single network, the route cost will be the maximum cost to any of the component subnets
  - **an inter-area path** is always composed of:
    - **an intra-area path** from the source to the abr
    - **a backbone path** between the source and destination areas
    - **an intra-area path** to the destination

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ospf path types

- **external paths** are learned from other routing protocols (e.g., bgp)
- **type 1**: the cost is expressed in terms of
  - **the external (bgp) route cost*** +
  - **the ospf cost** to the asbr

  example with bgp cost=495, ospf cost=10:

```
N E1 50.0.0.0/16        [505] tag: 0
                        via 10.0.1.2, eth1
```

* cost used when redistributing the protocol (bgp) into ospf; default for bgp=20; configurable by using `redistribute bgp metric` *`value`*

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ospf path types

- **type 2**: the cost is expressed in terms of
  - **the external (bgp) route cost* only** (distance to the asbr is only used <u>to break ties</u>)

example with bgp cost=495, ospf cost=10:

```
N E2 50.0.0.0/16      [10/495] tag: 0
                       via 10.0.1.2, eth1
```

  - metric type is user-configurable

```
redistribute bgp metric-type 2 metric 495
```

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experiments

- check that routers know detailed topology information only about their own area

r2

```
r2# show ip ospf neighbor

Neighbor ID     Pri State           Up Time    Dead Time Address  Interface      RXmtL RqstL DBsmL
200.0.0.1         1 Full/Backup     1m17s      32.279s 200.0.0.1  eth0:200.0.0.2     0     0     0
```

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experiments

- check that routers know detailed topology information only about their own area

r2

```
r2# show ip ospf database router
 OSPF Router with ID (200.0.1.1)
                Router Link States (Area 1.1.1.1 [Stub])
  ...
  Link State ID: 110.0.0.1
  ...
   Number of Links: 1
    Link connected to: a Transit Network
     (Link ID) Designated Router address: 100.0.0.2
     (Link Data) Router Interface address: 100.0.0.1
    ...
  Link State ID: 200.0.0.1
  ...
   Number of Links: 2
    Link connected to: a Transit Network
     (Link ID) Designated Router address: 100.0.0.2
     (Link Data) Router Interface address: 100.0.0.2
     ...
    Link connected to: a Transit Network
     (Link ID) Designated Router address: 200.0.0.2
     (Link Data) Router Interface address: 200.0.0.1
     ...
```

note: the output has been summarized

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experiments

- check that routers know detailed topology information only about their own area

r2

```
r2# show ip ospf database network
       OSPF Router with ID (200.0.1.1)
                Net Link States (Area 1.1.1.1 [Stub])
  LS age: 759
  Options: 0x0  : *|-|-|-|-|-|-|-
  LS Flags: 0x6
  LS Type: network-LSA
  Link State ID: 100.0.0.2 (address of Designated Router)
  Advertising Router: 200.0.0.1
  LS Seq Number: 80000001
  Checksum: 0x09ec
  Length: 32
  Network Mask: /30
        Attached Router: 110.0.0.1
        Attached Router: 200.0.0.1
  LS age: 763
  Options: 0x0  : *|-|-|-|-|-|-|-
  LS Flags: 0x3
  LS Type: network-LSA
  Link State ID: 200.0.0.2 (address of Designated Router)
  Advertising Router: 200.0.1.1
  LS Seq Number: 80000001
  ...
```

note: the output has been summarized

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experiments

- check what routers know about the outside of the area, using the **`show ip ospf database summary`** command
  - in particular, check the **`Metric`** values, that show how far away the destination is from the advertising abr
- check that routers in stub areas are offered a default route, whereas routers in the backbone are not
  - also check what **`Metric`** is assigned to the default route

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experiments

- experiment ospf's recovery capabilities
  - when multiple equal cost routes are available, ospf keeps all of them
  - check it by verifying what **`r3`** knows about the default route

r3

```
r3# show ip ospf route
============ OSPF network routing table ============
N IA 0.0.0.0/0              [11] area: 2.2.2.2
                            via 110.0.0.1, eth1
                            via 120.0.0.1, eth0

...
```

inter-area path

equal cost routes

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experiments

- experiment ospf's recovery capabilities
  - when multiple equal cost routes are available, ospf keeps all of them
  - check it by verifying what **`r3`** knows about the default route
  - zebra performs the actual selection

r3

```
r3# show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP, O - OSPF,
       I - IS-IS, B - BGP, E - EIGRP, N - NHRP, T - Table, v - VNC,
       V - VNC-Direct, A - Babel, F - PBR, f - OpenFabric, > - selected route,
       * - FIB route, q - queued, r - rejected, b - backup t - trapped, o - offload
failure

O>* 0.0.0.0/0 [110/11] via 110.0.0.1, eth1, weight 1, 00:18:15
  *                    via 120.0.0.1, eth0, weight 1, 00:18:15
```

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experiments

- experiment ospf's recovery capabilities
  - when multiple equal cost routes are available, ospf keeps all of them
  - check it by verifying what **`r3`** knows about the default route
  - zebra performs the actual selection
    - now bring **`bb1`**'s **`eth3`** down using **`ifconfig`**, wait a few seconds and check how the routing is changed
    - bring **`bb1`**'s **`eth3`** back up and check again how the routing is changed

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lab: ospf-complex

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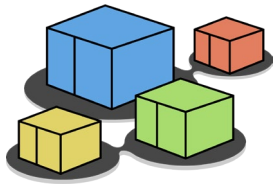

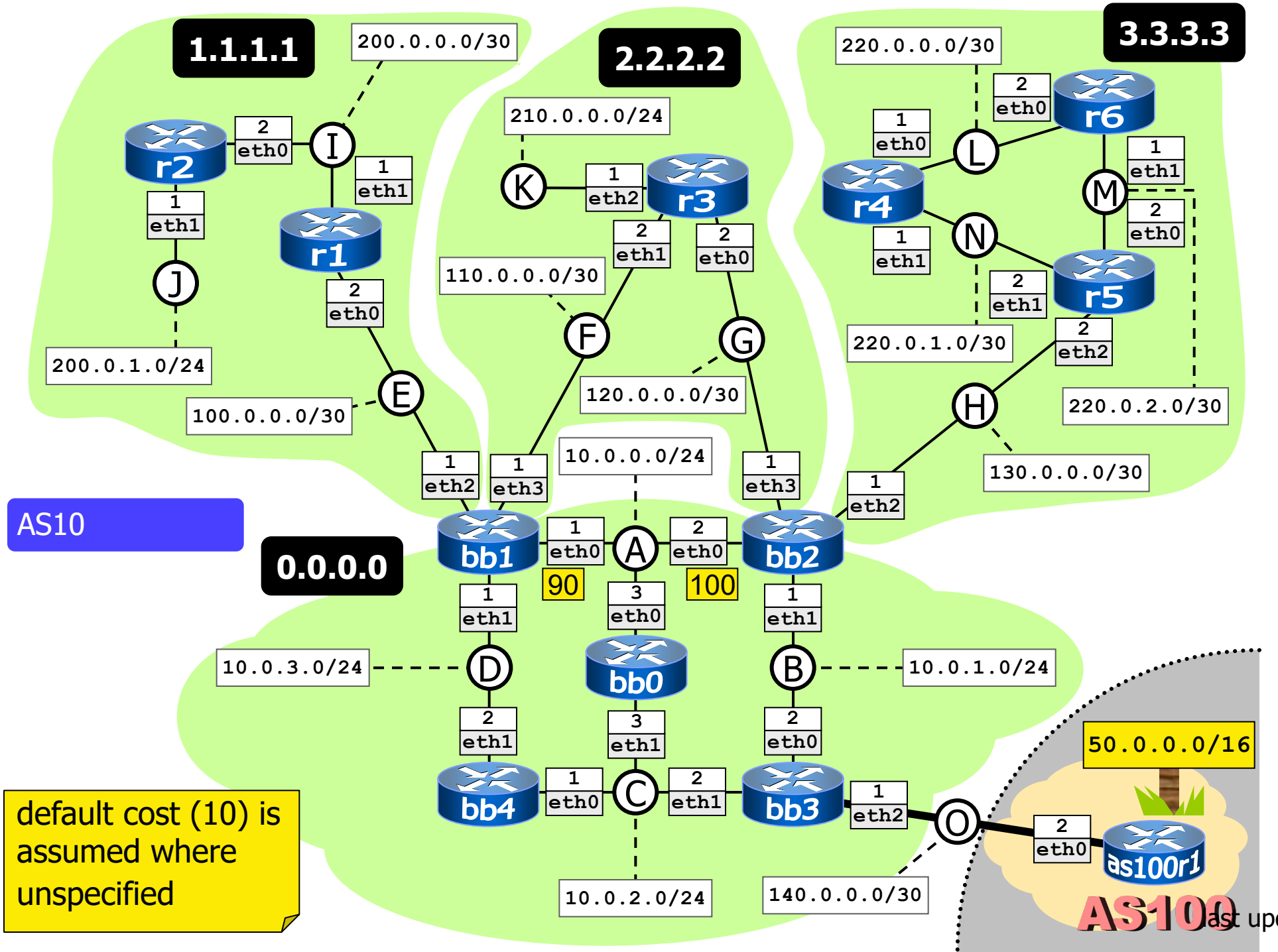

default cost (10) is assumed where unspecified

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lab description

- same as multiarea + some information is injected via bgp from an external as
  - also, abrs are configured to just inject the default route
    `area 1.1.1.1 stub no-summary`
- perform the same experiments as for the multiarea lab
  - in addition, check asbr information using
    - **`show ip ospf database asbr-summary`**
  - also check that such information is not propagated inside stub areas

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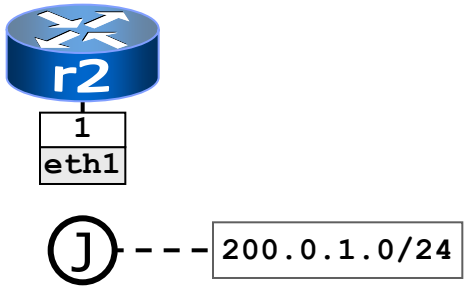
A quick note about stub networks

- "stub" = not used for transit
- three possible situations:



ospf is enabled also on the stub network's interface → then → the stub network is advertised in the entire ospf domain → but → interface uselessly sends hello packets

ospf is **not** enabled on the stub network's interface → then → the stub network is advertised → but → the stub network is **advertised as an AS-external route** (hence, only through the backbone)

ospf is enabled on the stub network's interface, which is **configured as** `passive-interface` ✓ → then → the stub network is advertised in the entire ospf domain → but → N/A

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